

Design Exercise (10%)

Design a network for a medium sized manufacturer based on the following specification:

Specification

The manufacturer has 200 staffs and spread across six buildings in one site.

- Production (100 staffs)
- Warehouse (20 staffs)
- Sales and Marketing (20 staffs)
- Finance (10 staffs)
- R&D (30 staffs)
- Management (20 staffs)

Each staff has dedicated computer connection point to connect their company desktop computer and/or laptops. In addition, some staffs (e.g. sales and marketing staffs) need Bring Your Own Device (BOYD) connection points. The manufacturer plans to expand in the near future.

Requirement

Design a comprehensive network to facilitate the following:

- Internal connectivity (IP address scheme, VLAN, etc)
- Layer 2 security
- Wireless LAN for mobile stations (laptop, mobile machineries)
- Access to the Internet
- Routing between internal and external networks
- Remote access
- Overall network security

Submission

Use Packet Tracer to design your network.

Write a design report to detail your design, including justification of the device choices and rational of the design, discussion, etc.

Word Limit: 2000 words

Submission Deadline

Submit via Moodle by Friday 5pm Week 10.

Marking Scheme

Item	Mark	Details	
Network Topology & Design	20%	Logical, clear network layout showing interconnection of buildings and devices. Use of appropriate topology	
IP Addressing & VLAN Configuration	10%	Correct use of subnetting, IP addressing scheme, VLAN segmentation based on departments or roles. Consideration for scalability and efficiency	
Wireless LAN & BYOD Support	10%	Proper inclusion of wireless access points, SSIDs, VLANs for BYOD, and appropriate coverage for mobile users and machinery	
Layer 2 Security	10%	Implementation of MAC filtering, port security, disabling unused ports and other techniques for secure the network at Layer 2	
Routing & Internet Access	5%	Clear method of routing between internal VLANs and Internet (e.g. static routing or basic dynamic protocols)	
Overall Network Security	10%	Consideration of firewalls, physical security, user authentication, antivirus/malware protection, etc	
Design Justification & Report Quality	20%	Quality of report. Logical explanation of choices (devices, topology, protocols), reflection on scalability, performance, and cost.	
Packet Tracer Implementation	10%	Network implemented in Cisco Packet Tracer according to design. Shows functionality (ping tests, routing table, VLAN assignment).	
Presentation & Documentation	5%	Clear structure, spelling, grammar, use of diagrams/screenshots, stays within 2000-word limit	